# Avoiding sillies

Michael Luo

October 18, 2026

# A motivating quote

*To succeed in competition math it is not enough to be right. It is also necessary to not be wrong.*

# What is a silly?

**Definition**

A *silly* is an almost-correct solution with a "dumb" mistake.

Sillies are Minnetonka's downfall.

# Examples

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- Arithmetic mistakes
- Forgetting edge cases
- Misreading the question
- Sign errors

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- Arithmetic mistakes
- Forgetting edge cases
- Misreading the question
- Sign errors
- Listening to a song with a number and writing that instead

# Non-examples

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Significant mistakes are not sillies. If there are $n$ problems and you scored $m$, you didn't silly $n - m$ problems.

- Fundamentally flawed approach
- Failed guess

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- Fundamentally flawed approach
- Failed guess
- Answer sheet spontaneously catches fire
- Accidentally eat chocolate containing alcohol before the contest

# The problem

Sillies are easily correctable.

**Corollary**

*Reducing sillies is an easy way to improve.*

Example: Me at HMMT 2025

# Factors increasing sillies

- Doing problems quickly (Me at ARML tiebreaks 2025)
- Headsolving or solving with low scratch paper
- Sleep deprivation or jet lag
- Strong, distracting emotions, such as excitement, anger, or heartbreak
- Sickness

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- Doing problems under the influence (don't drink and derive)
- Having family history of sillying

# Factors ~~increasing~~ decreasing sillies

- Doing problems ~~quickly~~ slowly
- ~~Headsolving or solving with low scratch paper~~ using scratch paper liberally
- ~~Strong, distracting emotions, such as excitement, anger, or heartbreak~~ being locked in

# Most important advice

Reasoning must be PAINFULLY explicit. Smart people make stupid mistakes. Write down all reasoning. Do not skip steps in your head!

Slow is smooth, smooth is fast. Sillies when barying ISL 2024 G4 turned 20 minutes into 3 hours.

# Pointing-and-calling

- In many Asian railways, conductors are required to literally point and call out what they're seeing and doing
- Sounds dumb? Shown to reduce error by 85%
- Being painfully explicit reduces errors

# The pointing-and-calling method




Figure: The process of pointing and calling. From Wikipedia.

# Good and bad scratchwork

Scratchwork isn't only for computation. Work out small examples. Write down reasoning.

Writing everything down is pointing-and-calling. If you think $2 \times 3$, don't write 6. Don't skip steps!

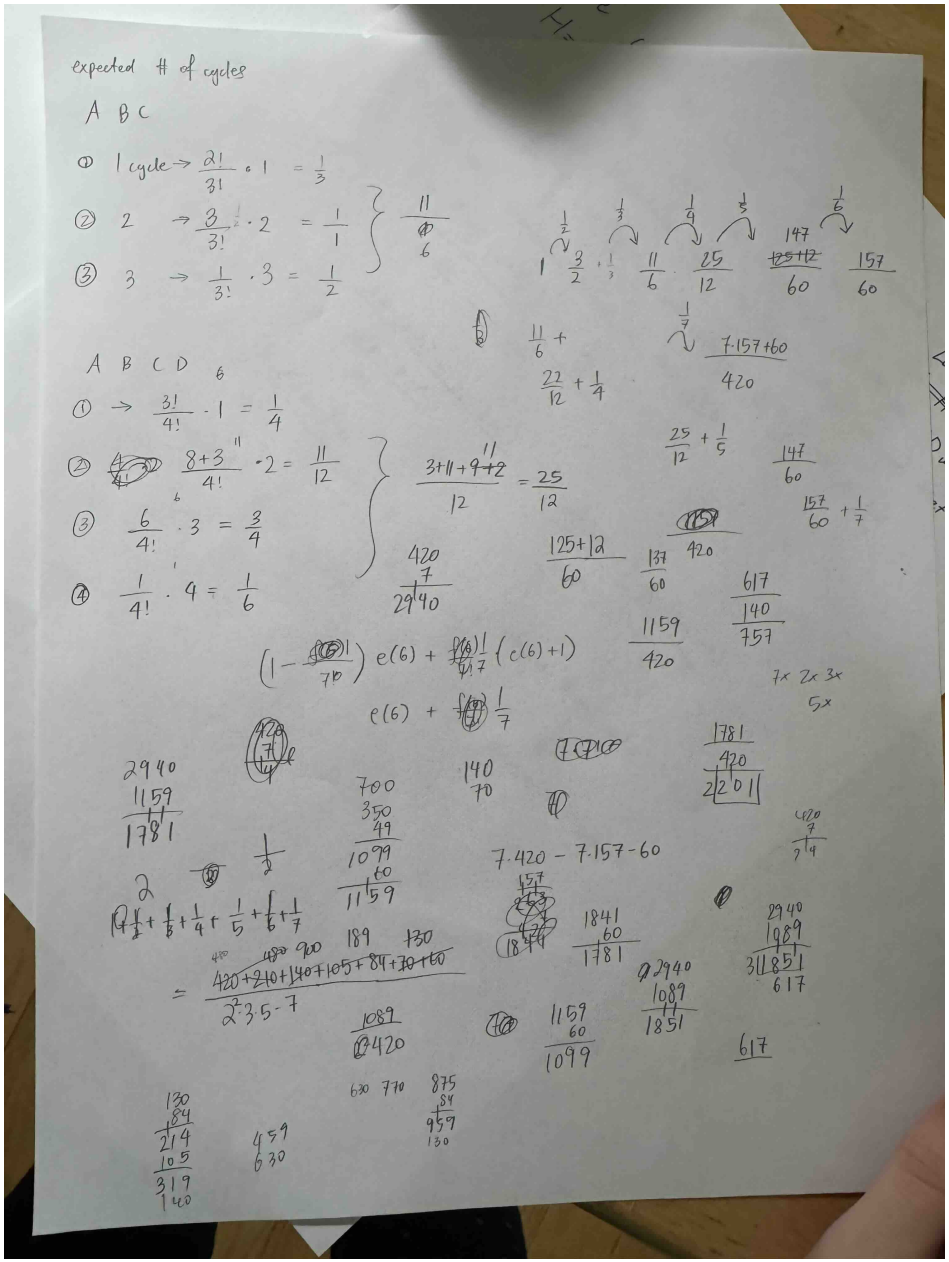
Don't use test paper margins (not enough space). You'll write over yourself and problems. You are not Fermat.

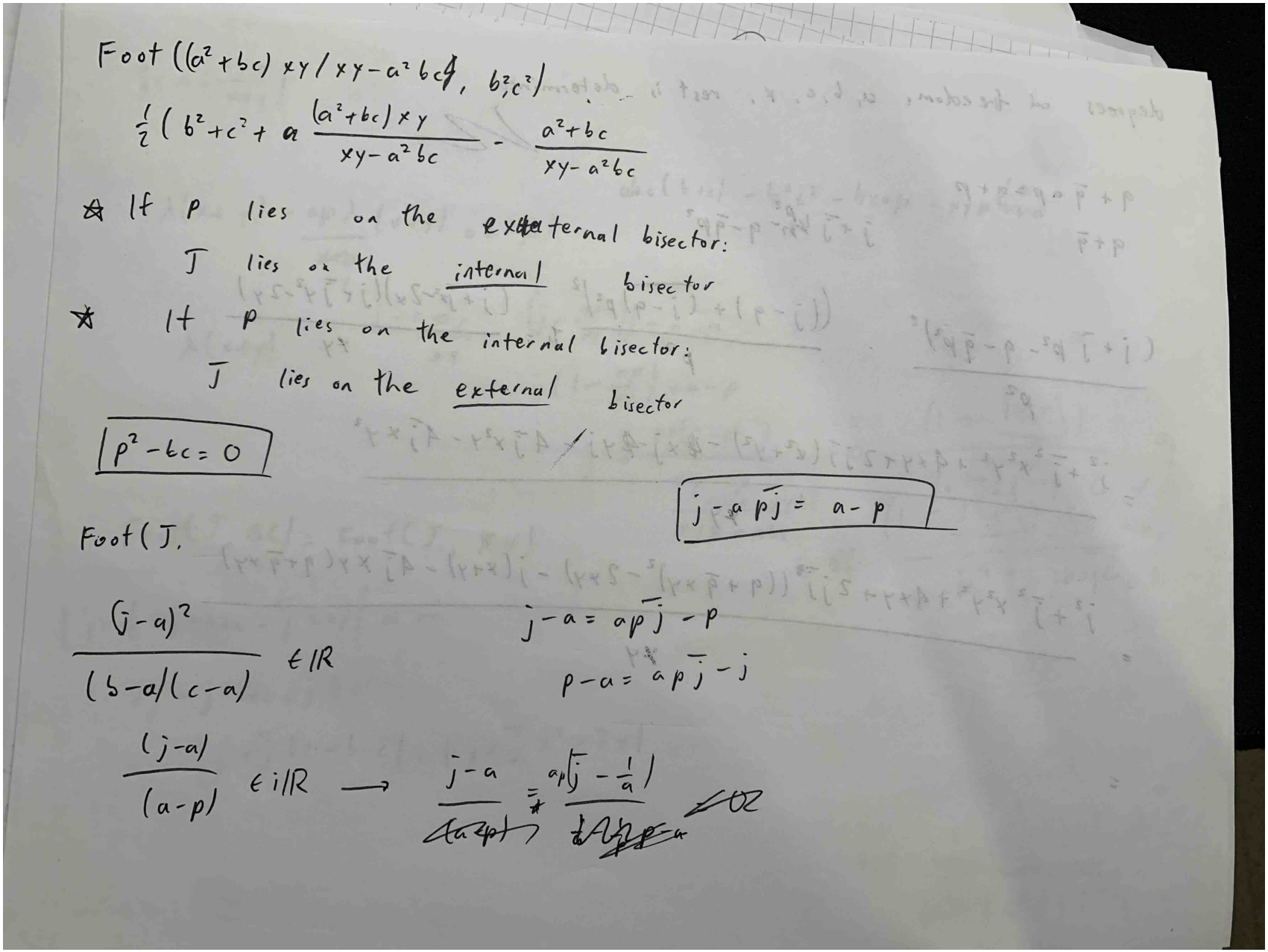
# Good and bad scratchwork

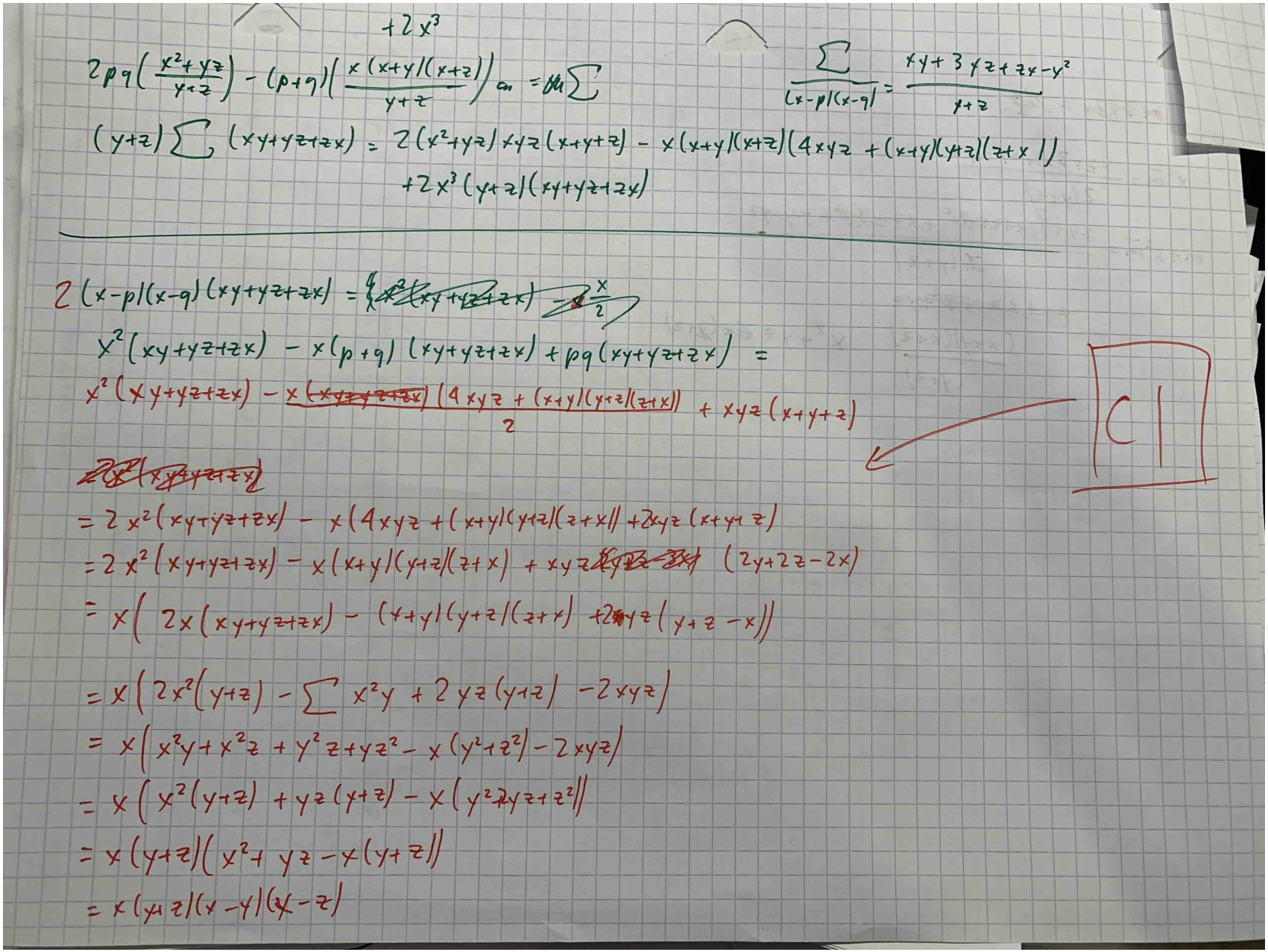
The following slides include examples of scratch paper.

Think about how their paper is organized and how work is being done.

# Lanie Deng




Foot((a²+bc) xy / xy − a²bc, b², c²)
½ ( b² + c² + a (a²+bc) xy / (xy − a²bc) − (a²+bc) / (xy − a²bc) )
★ If P lies on the external bisector:
J lies on the internal bisector
★ If P lies on the internal bisector:
J lies on the external bisector
p² − bc = 0
j − a p̄ j = a − p
Foot(J,
(j−a)² / ((b−a)/(c−a)) ∈ iℝ
j − a = a p̄ j − p
p − a = a p̄ j − j
(j−a) / (a−p) ∈ iℝ ⟶ j − a / a(p̄ j − 1/a) = ★


C1

- if $2xy$ is a perfect square!

$x = \frac{44^2}{2}$: 20, 21, 22, 23 work

$x + \log_8(y) = 1024 \rightarrow$

$y = 8^{1024-x} \sim 2^{1024-x}$

- if $x$ is even, then $2xy = \sim 2x \cdot$ ~~$2^{1024-x}$~~ is a square

- if $x$ is odd, then $2xy \sim 2x 2^{1024-x} \sim x$ is a square

$1^2, 3^2, 5^2, \ldots,$ ~~$31^2$~~ $< 1024$ $\quad 16 \times 22 = 38$

$\frac{2^2}{2}, \frac{4^2}{2}, \ldots, \frac{44^2}{2}$ $\quad \left(\frac{45^2}{2}\right) > 1024$

$46^2 > 2048$

$N$, $N+2019$ are palindromes

The last digit goes down by 1

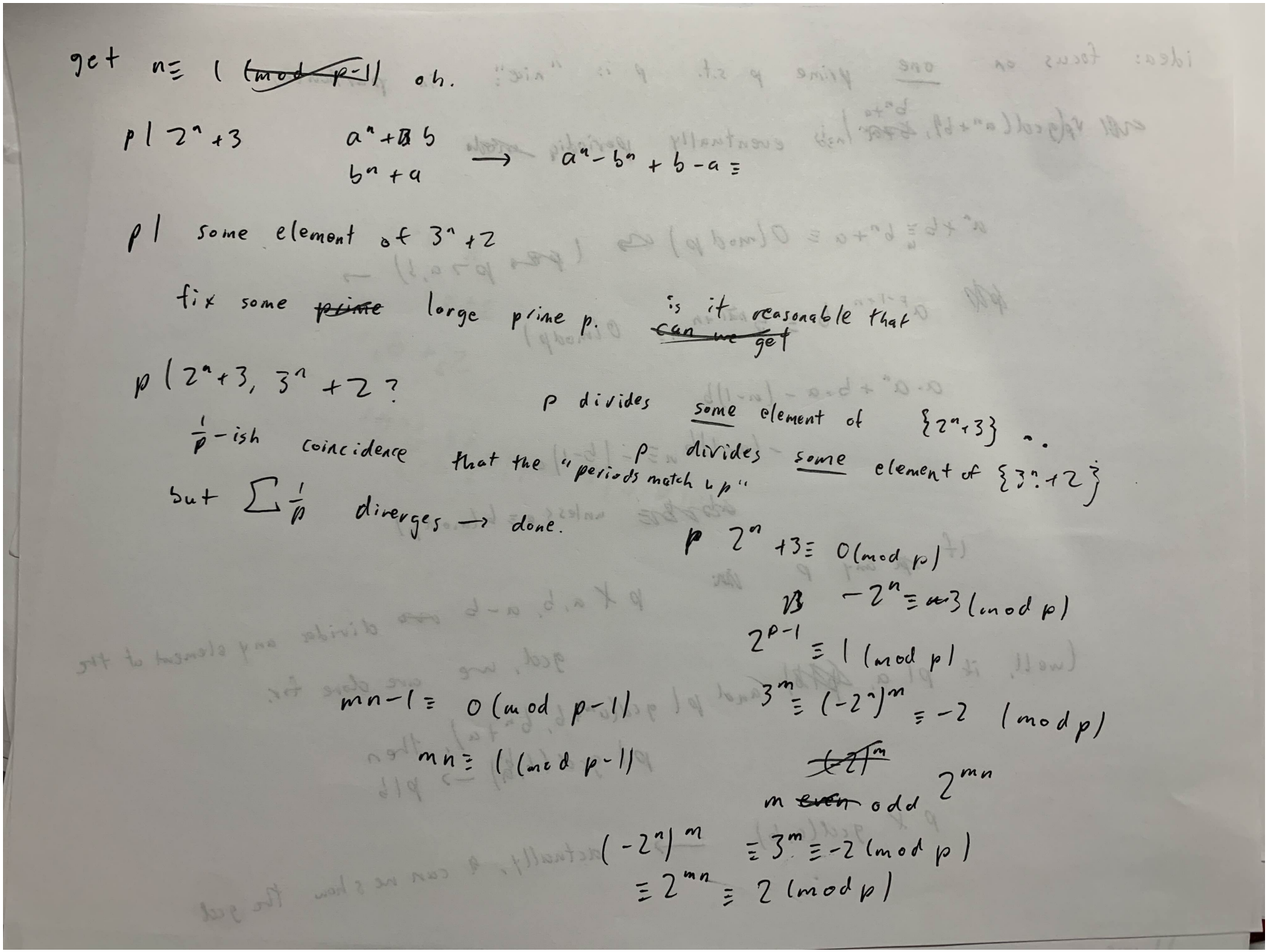
$$\begin{array}{r} \underline{a\ b\ c} \quad \underline{c\ b\ a} \\ 2 \quad 0\ 1\ 9 \end{array}$$

2019

(last digit can't be 0) $\quad 1 \rightarrow 0$, etc.

hence there needs to be an "overfill"

(in particular, we want $2 \rightarrow 1$)

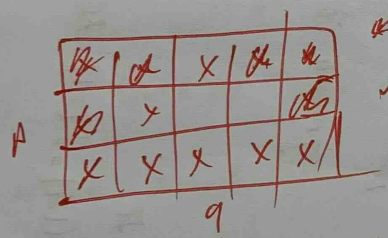

get $n \equiv 1 \pmod{p-1}$ oh.
$p \mid 2^n+3$ $a^n+b^n$ $b^n+a$ $\longrightarrow$ $a^n-b^n+b-a \equiv$
$p \mid$ some element of $3^n+2$
fix some ~~prime~~ large prime $p$. is it reasonable that ~~can we~~ get
$p \mid 2^n+3,\ 3^n+2$?
$p$ divides some element of $\{2^n+3\}$ ..
$\frac{1}{p}$-ish coincidence that the "periods match up" $p$ divides some element of $\{3^n+2\}$
but $\sum \frac{1}{p}$ diverges $\longrightarrow$ done.
$p \mid 2^n+3 \equiv 0 \pmod p$
$-2^n \equiv 3 \pmod p$
$2^{p-1} \equiv 1 \pmod p$
$3^m \equiv (-2^n)^m \equiv -2 \pmod p$
$mn-1 \equiv 0 \pmod{p-1}$
$mn \equiv 1 \pmod{p-1}$
~~$(-2)^m$~~ $2^{mn}$
$m$ ~~even~~ odd
$(-2^n)^m \equiv 3^m \equiv -2 \pmod p$
$\equiv 2^{mn} \equiv 2 \pmod p$

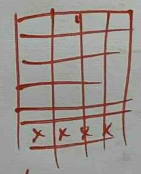
# Michael Luo

- The set must be surjective modulo all but at most one prime.
  (Proof. If else, then [illegible] that cross-section, it's ambiguous)
- The graph must be connected.

Answer: $2 \cdot 89 \cdot 97 - 1$.

¶ Construction: Pick everything to be 0 mod 97. Then just make the grid mod $pq$ for all else. look like this.

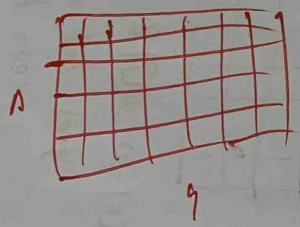


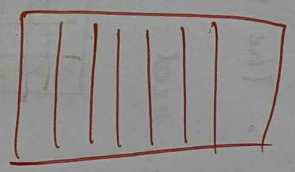

$p+q-1$

This works — it detects periodicity mod $q$ always.

¶ Bound. 87, 89, and 97, pick one,




$2 \cdot 87 - 1$ is the bound



$89+87-1 = 175$

$$\begin{array}{r} 88 \\ \times\ 2 \\ \hline 16 \\ \times\ 10 \\ \hline 175. \end{array}$$

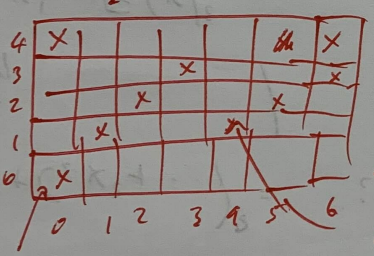
$p_1, \dots, p_n$ are the primes $< 100$.

we need to find (for all but 1)

$i \neq j$ s.t. $f(i) \neq f(j)$ in $\mathbb{Z}/(q)$

$2 \cdot 89 = 178$ works, yet? The bound is not good — we can make it ambiguous between 89 and 97. Is the answer smaller?

Can we choose $2 \cdot 89 - 2$ ? $\pmod 5$




like

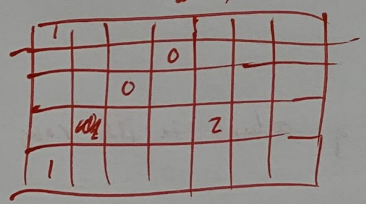
choose $q \mid 10$ such that $\pmod 7$

on its own

oh. we are playing "perfectly"

If we only pick $q$, what goes wrong?

Bob:



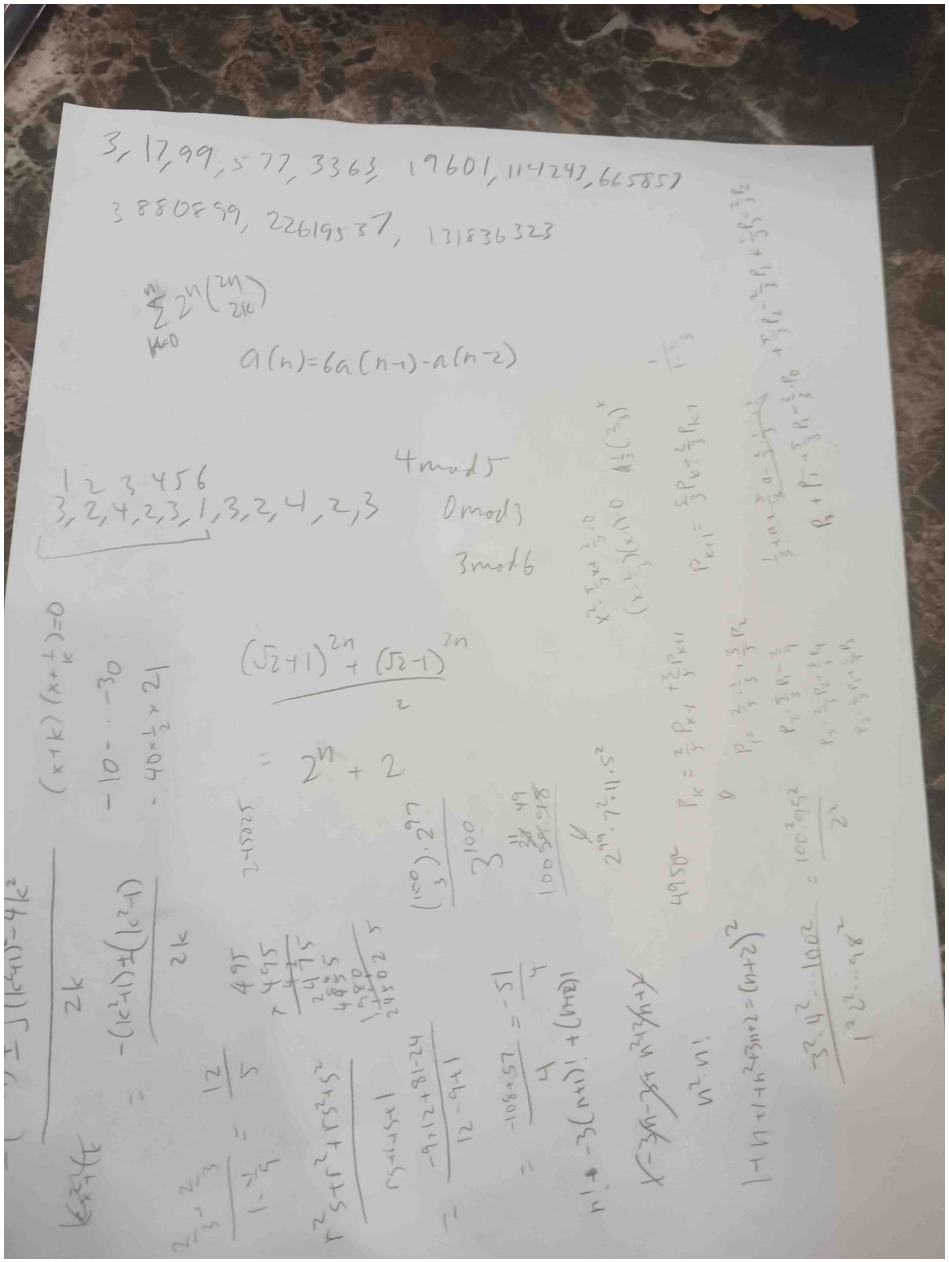

element on its own

select $2q-1$ squares such that it is impossible to tell them s.t. ambiguity if two elements are

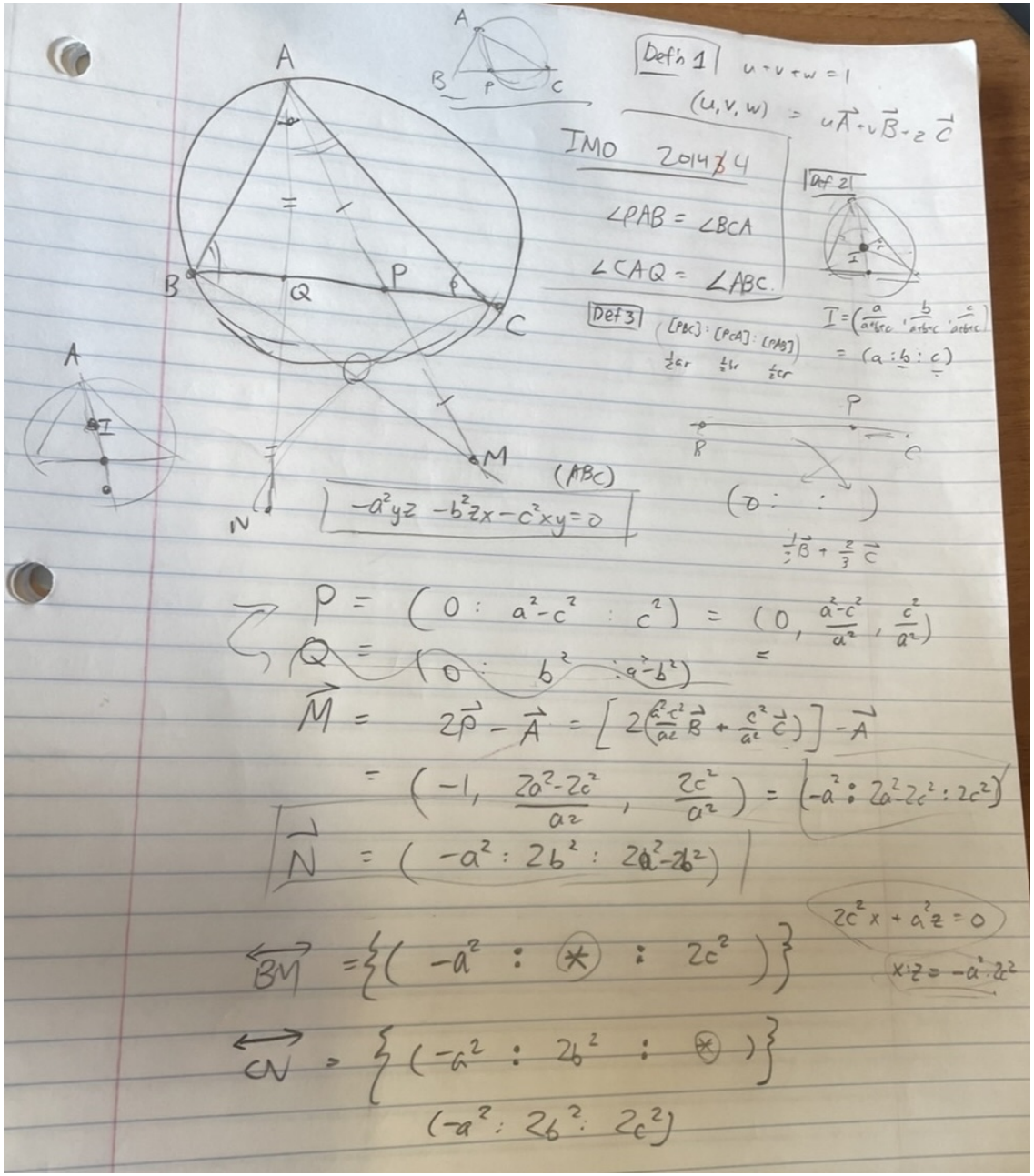
if two elements in same row, same value

or two elements in same column, same value $\rightarrow q$.

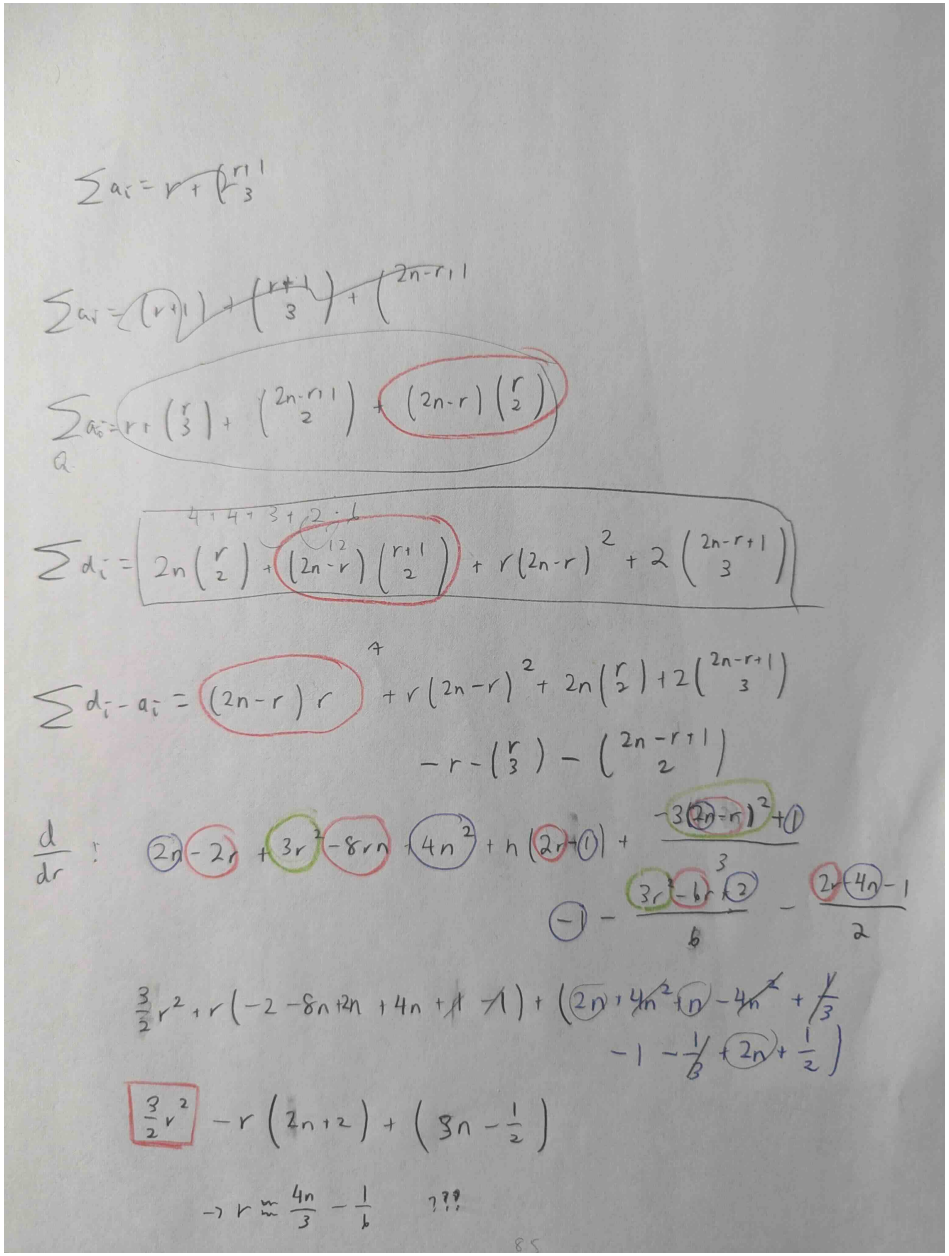
# Selena Ge


3, 17, 99, 577, 3363, 19601, 114243, 665857
3880899, 22619537, 131836323
a(n)=6a(n-1)-a(n-2)
4 mod 5
0 mod 3
3 mod 6
3, 2, 4, 2, 3, 1, 3, 2, 4, 2, 3
$2^n + 2$

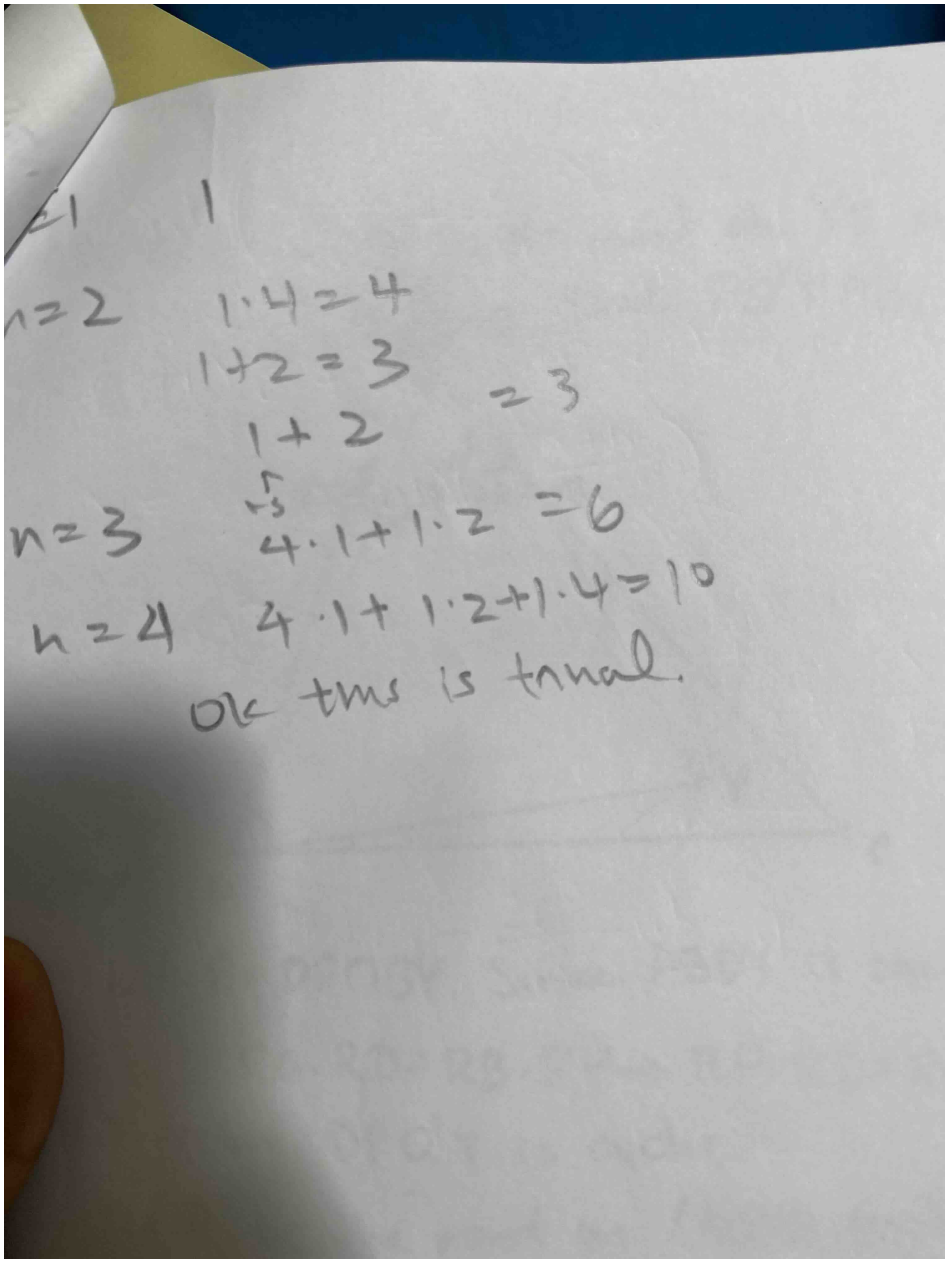
# Evan Chen


A
B
C
P
Q
M
N
I
Def 1
u + v + w = 1
(u, v, w) = uA + vB + wC
IMO 2014/4
∠PAB = ∠BCA
∠CAQ = ∠ABC
Def 2
Def 3
[PBC] : [PCA] : [PAB]
1st 2nd 3rd
I = (a/(a+b+c), b/(a+b+c), c/(a+b+c))
= (a : b : c)
(ABC)
−a²yz − b²zx − c²xy = 0
P = (0 : a² − c² : c²) = (0, (a²−c²)/a², c²/a²)
Q = (0 : b² : a² − b²)
M = 2P − A = [2((a²−c²)/a² B + c²/a² C)] − A
= (−1, (2a²−2c²)/a², 2c²/a²) = (−a² : 2a²−2c² : 2c²)
N = (−a² : 2b² : 2a²−2b²)
BM = {(−a² : ✱ : 2c²)}
CN = {(−a² : 2b² : ✱)}
(−a² : 2b² : 2c²)
2c²x + a²z = 0
x : z = −a² : 2c²

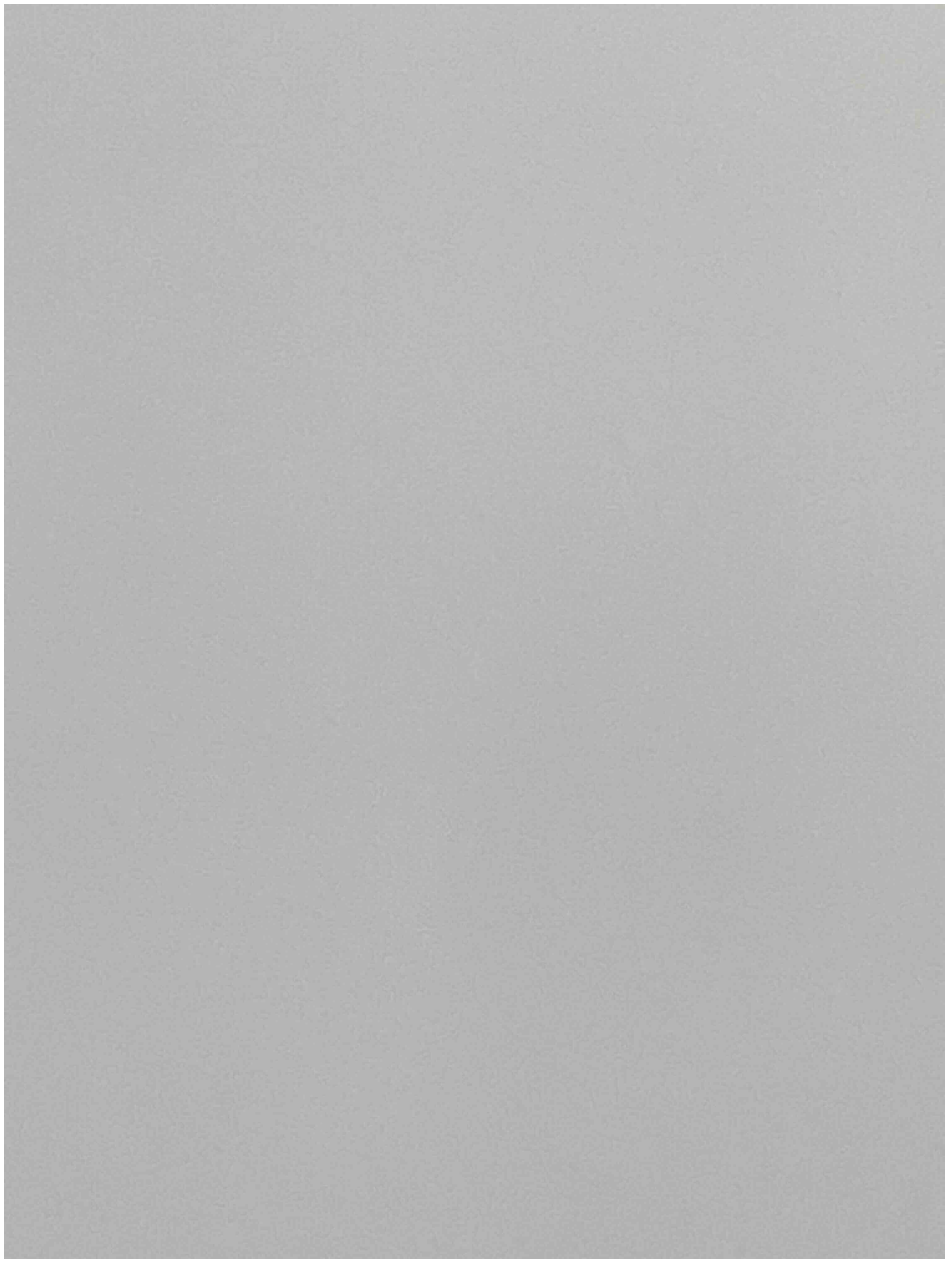
Jiahe Liu



Andrew Lin



# Alexander Wang



# Practice meet today

We'll do Sample Meet 4 Coyote. Write down reasoning in words. I'll check! If you do not do this, you will face execution.

Good luck!